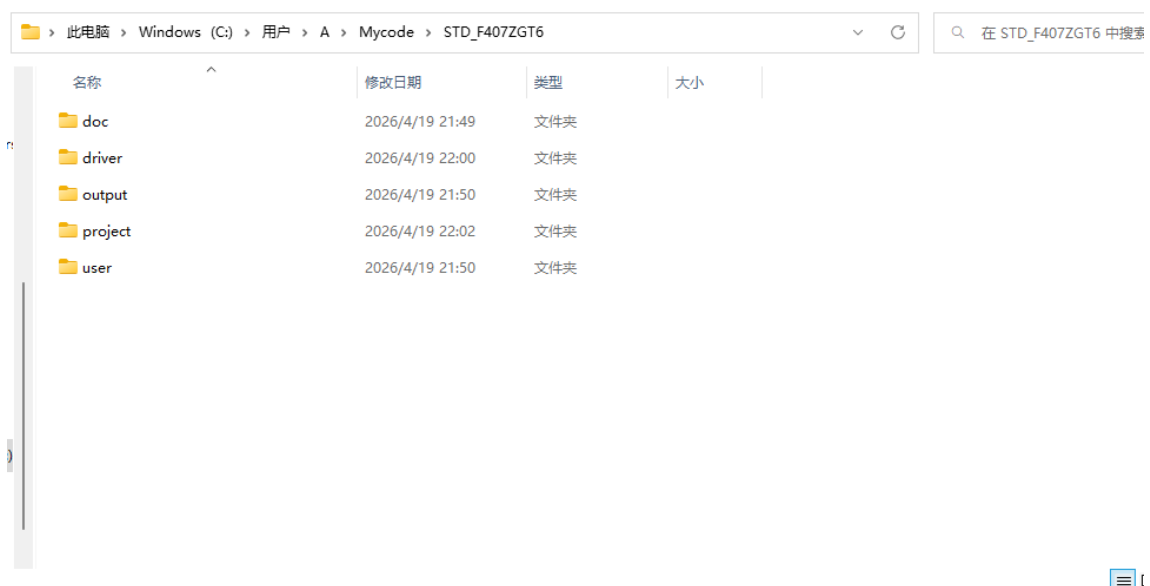
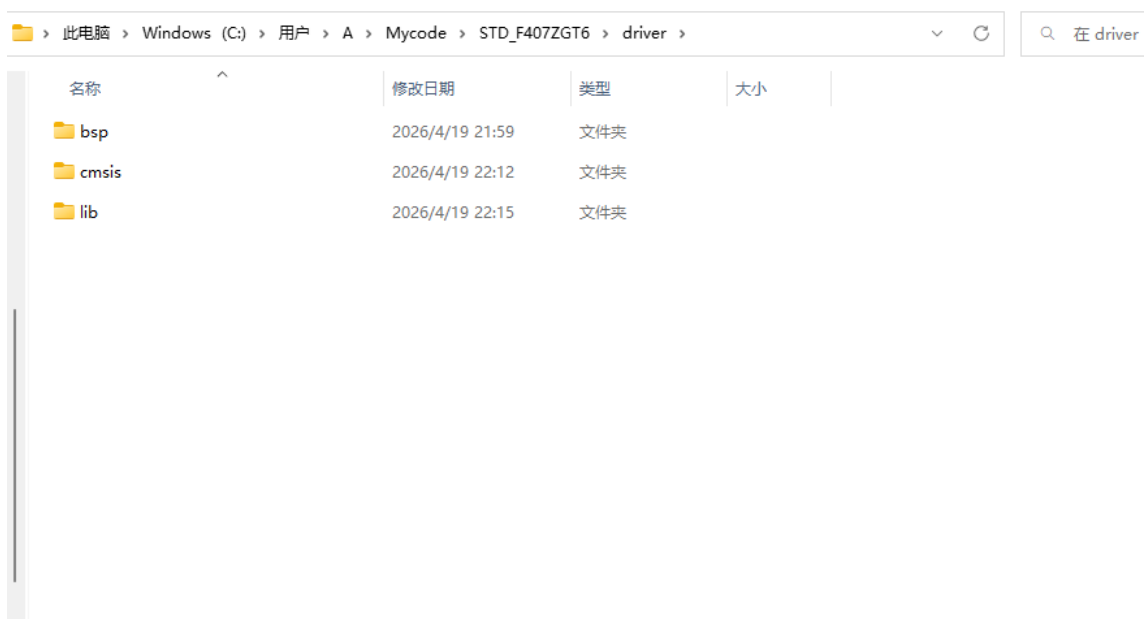


1. 创建总文件夹, 里面创建 `project`, `output`, `user`, `driver`, `doc` (比赛的时候不用)

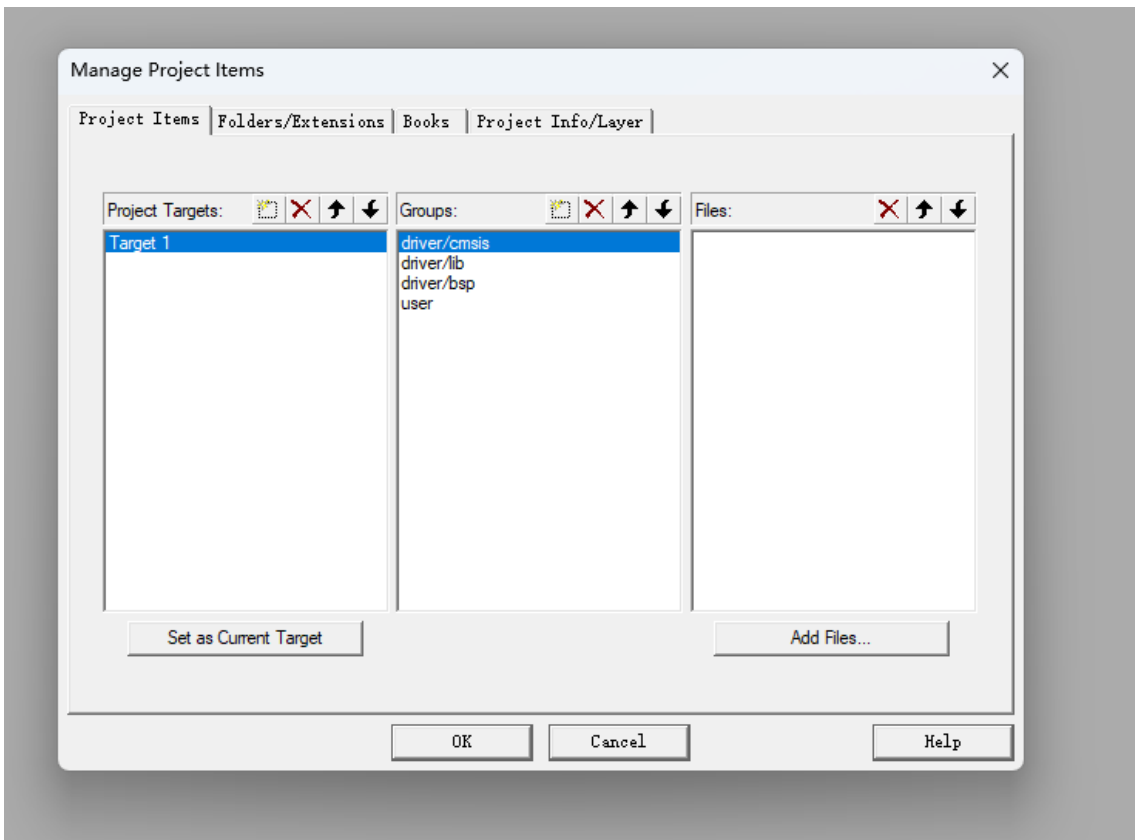


2. 在 `driver` 里面创建 `cmsis`, `bsp`, `lib`, 分别表示固件, 模块驱动, 标准库驱动



夹, 以及 C:\Users\A\Desktop\2026B平台资料2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Project\STM32F4xx_StdPeriph

3. 打开keil, 新建工程, 位置选择在 `project` 下, 选择芯片
4. 添加固件: 在keil里面配置文件夹, 在group下面新建 `driver/cmsis`, `driver/lib`, `driver/bsp`, `user` 这几个组



5. 在 `C:\Users\A\Desktop\2026B平台资料`
`2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Libraries\CMSIS\Device\ST\STM32F4xx\Source\`
`Templates\arm` 找到文件 `startup_stm32f40_41xxx.s`

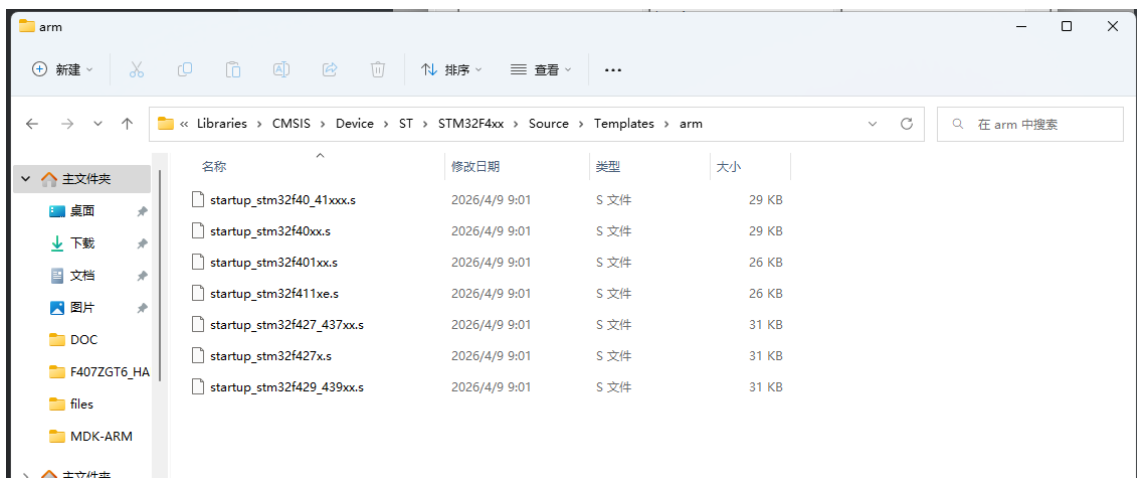
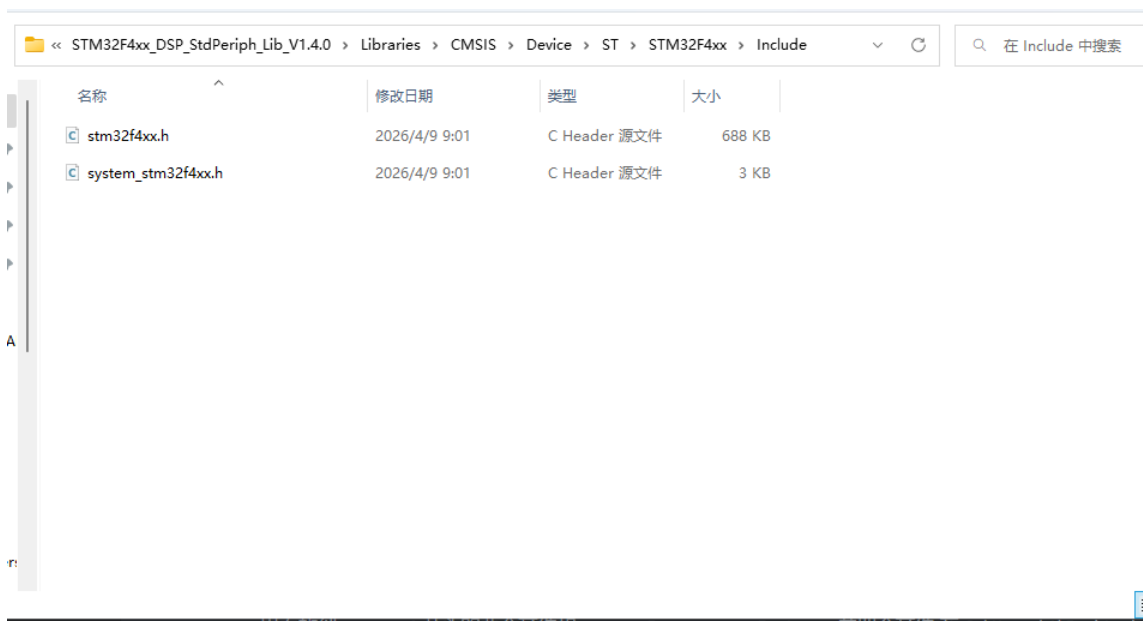


Figure 1

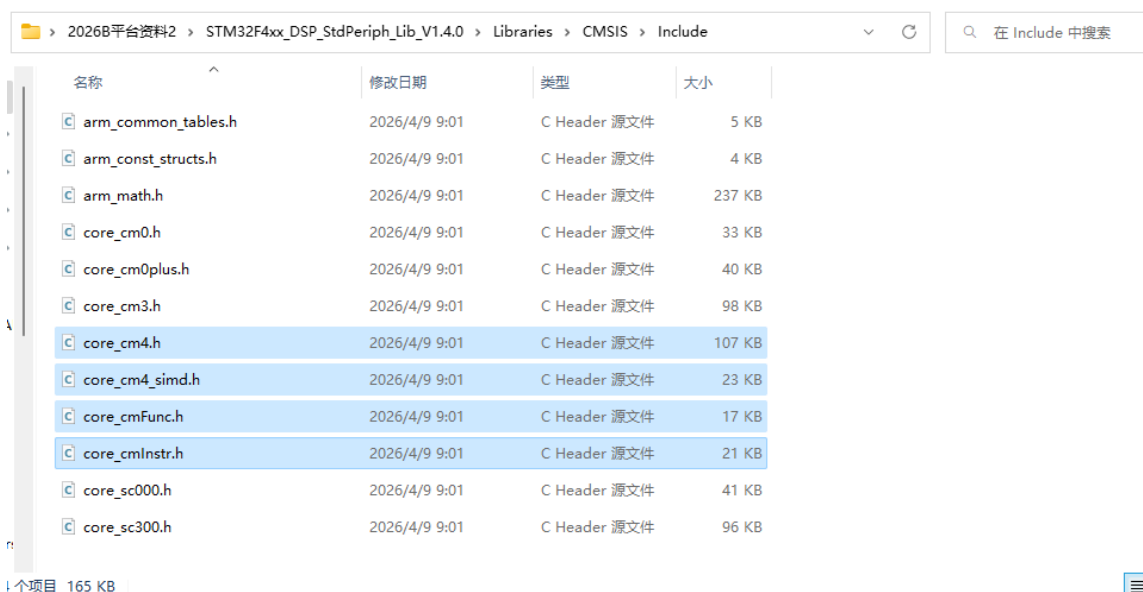
在 `C:\Users\A\Desktop\2026B平台资料`
`2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Libraries\CMSIS\Device\ST\STM32F4xx\Include`
 找到两个文件



在 C:\Users\A\Desktop\2026B平台资料

2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Libraries\CMSIS\Include

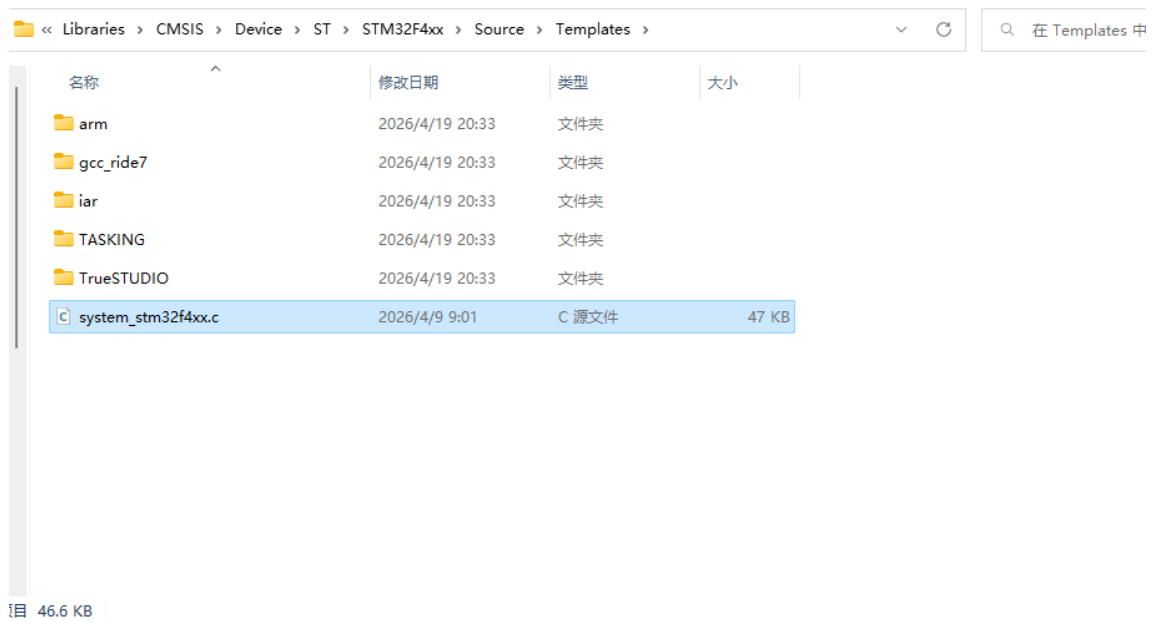
里面找到 core_cm4 开头的两个文件和 core_cmFunc core_cmInstr 一共四个文件



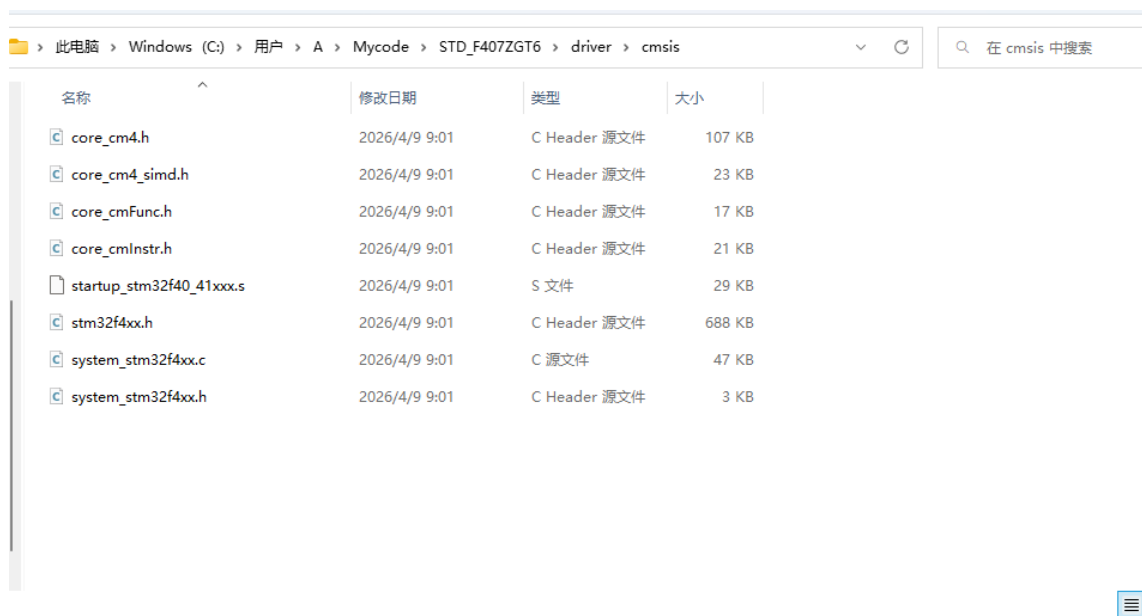
在 C:\Users\A\Desktop\2026B平台资料

2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Libraries\CMSIS\Device\ST\STM32F4xx\Source\Templates

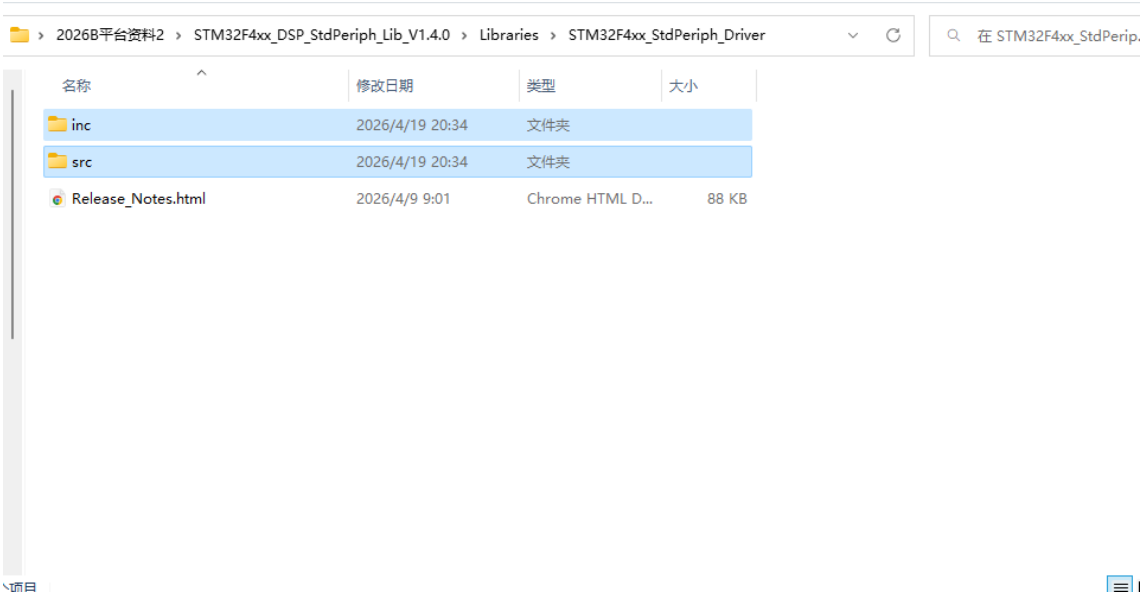
找到 system_stm32f4xx.c;



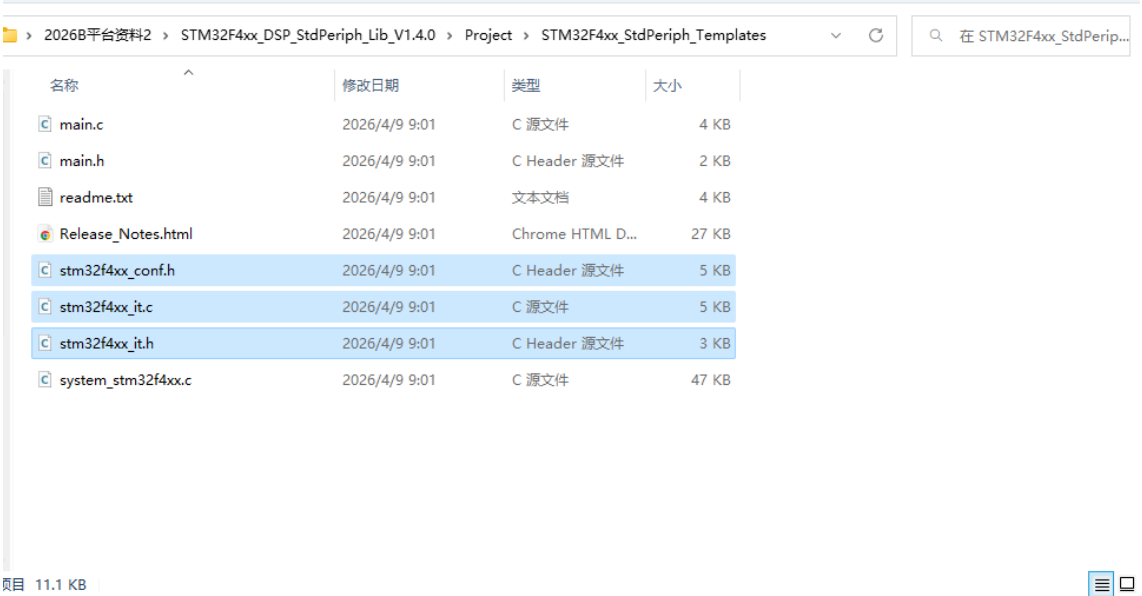
一共8个文件,全部移动到 cmsis 目录下。



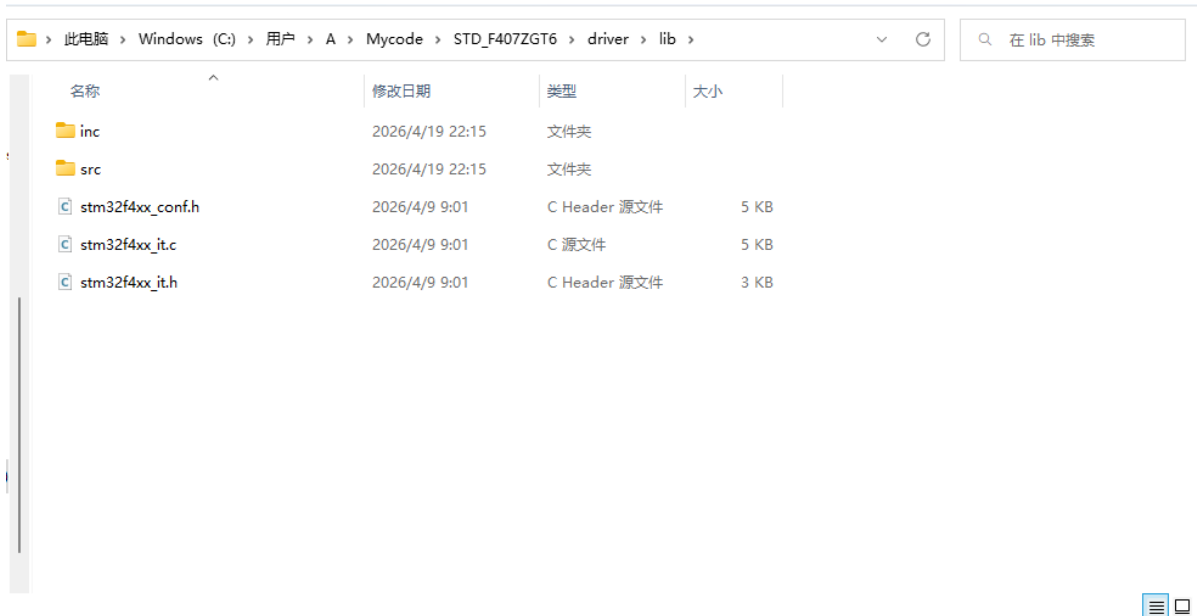
6. 添加库函数本尊: 在 C:\Users\A\Desktop\2026B平台资料2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Libraries\STM32F4xx_StdPeriph_Driver 找到inc和src两个文件夹



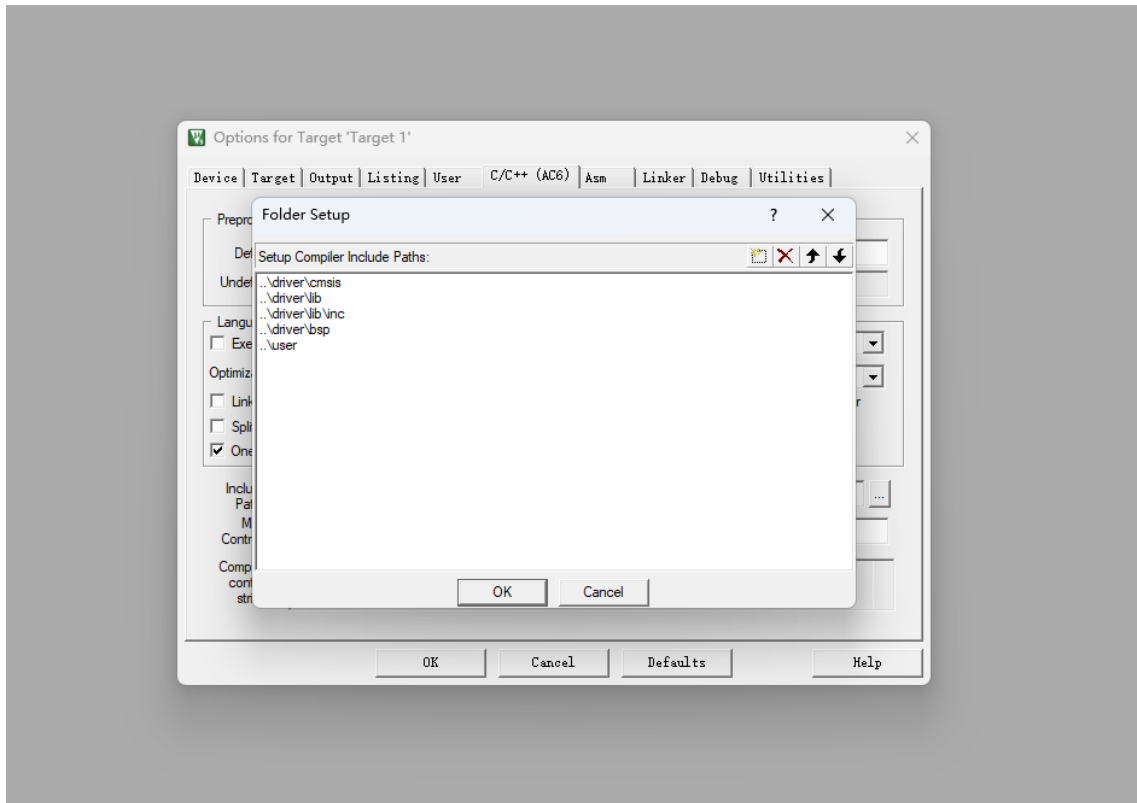
以及 C:\Users\A\Desktop\2026B平台资料
2\STM32F4xx_DSP_StdPeriph_Lib_V1.4.0\Project\STM32F4xx_StdPeriph_Templates 下找到
这三个文件



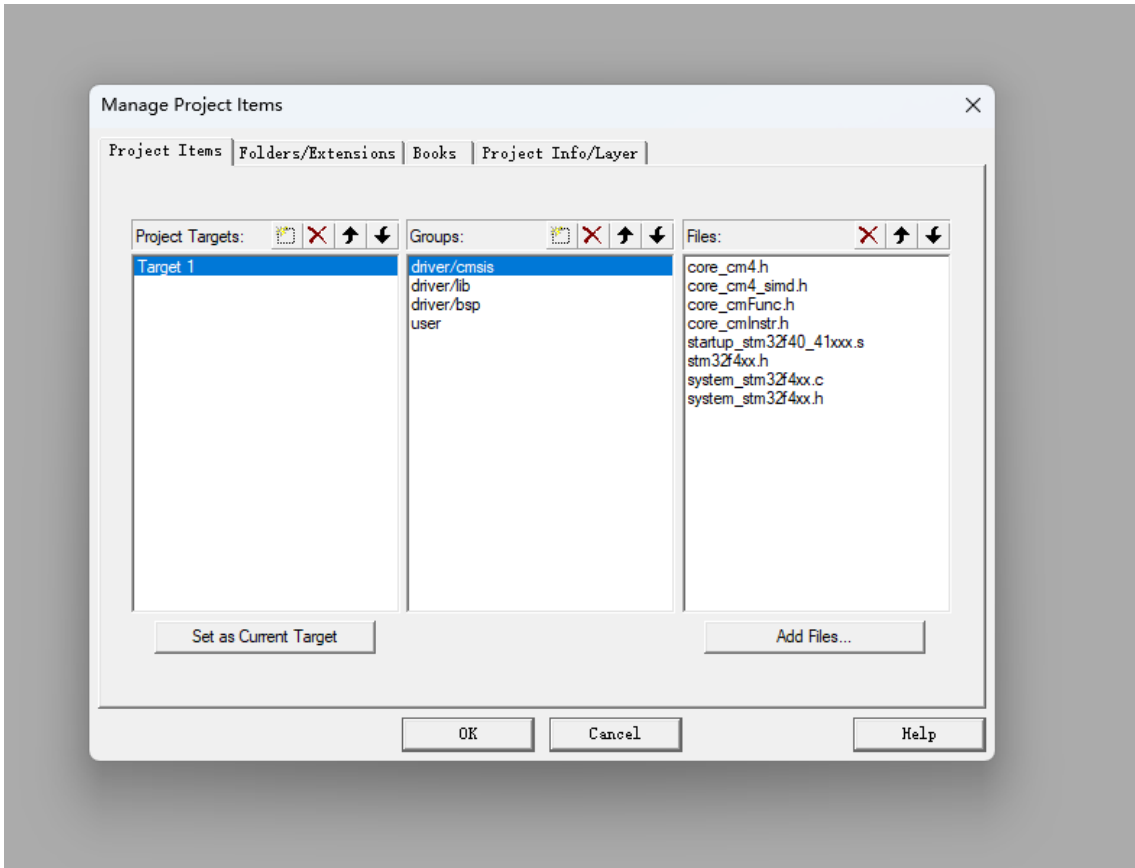
将他们移动到 driver/lib 下面



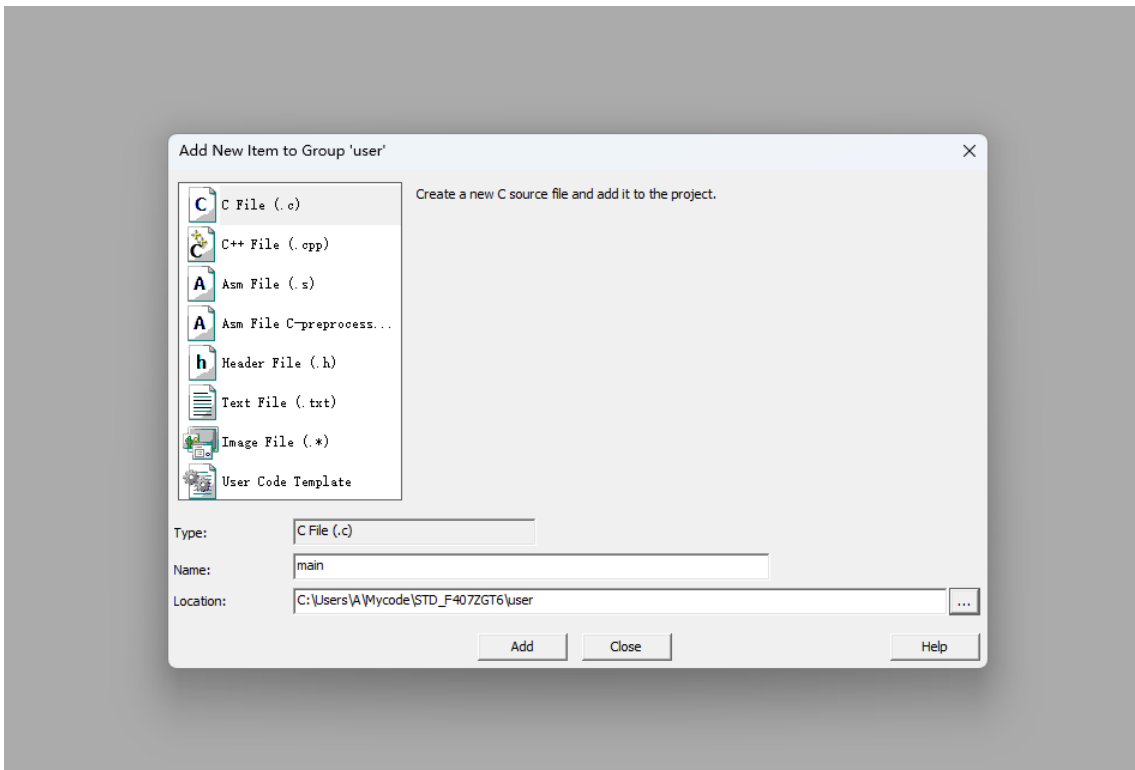
7. 在keil里面添加头文件路径



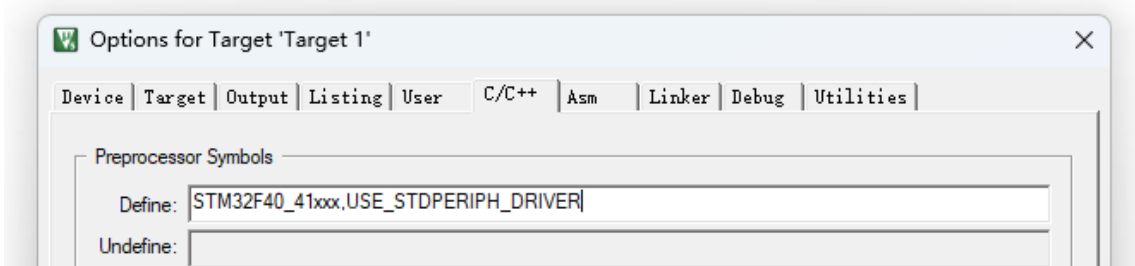
8. 在keil里面依次把之前提到的文件的本体都加进来



9. 在user下加一个main.c,注意文件的路径需要在项目的user/下面



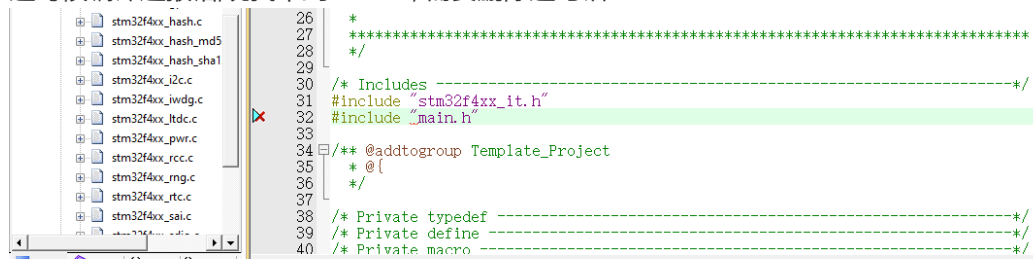
10. 在keil里面定义芯片宏和标准库宏



11. 这时候编译可能报错误,这是由于有些功能在这个芯片上用不了,需要移除相关文件

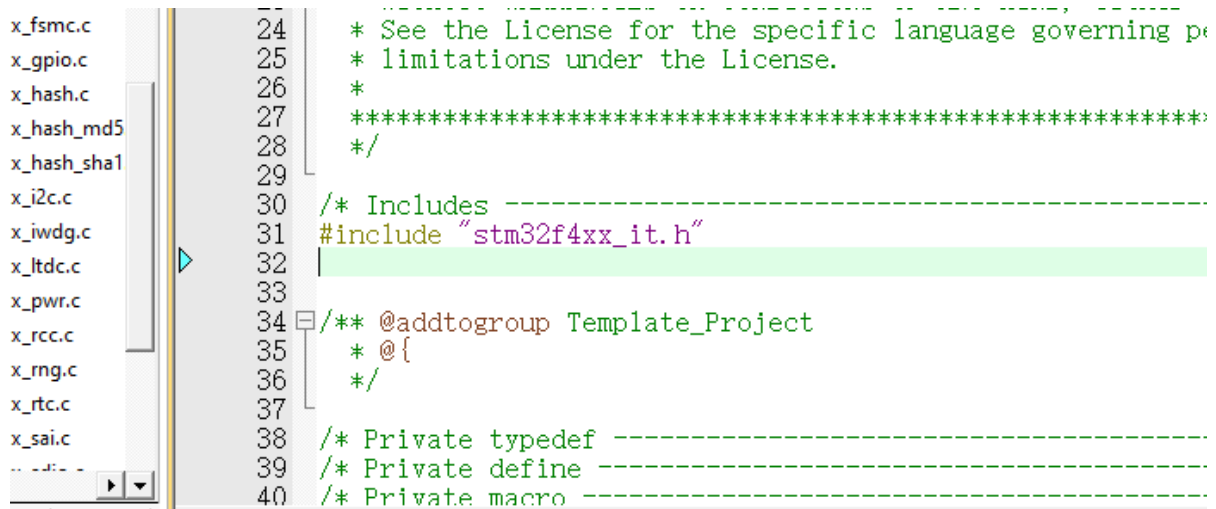
```
Build Output
.\driver\lib\src\stm32f4xx_fmc.c(640): error: #20: identifier "FMC_Bank4" is undefined
FMC_Bank4->PCR4 = 0x00000018;
.\driver\lib\src\stm32f4xx_fmc.c(676): error: #20: identifier "FMC_Bank4" is undefined
FMC_Bank4->PCR4 = (uint32_t)FMC_PCCARDInitStruct->FMC_Waitfeature |
.\driver\lib\src\stm32f4xx_fmc.c(739): error: #20: identifier "FMC_Bank4" is undefined
FMC_Bank4->PCR4 |= PCR_PBKEN_SET;
.\driver\lib\src\stm32f4xx_fmc.c(744): error: #20: identifier "FMC_Bank4" is undefined
FMC_Bank4->PCR4 &= PCR_PBKEN_RESET;
.\driver\lib\src\stm32f4xx_fmc.c(809): error: #20: identifier "FMC_Bank5_6" is undefined
FMC_Bank5_6->SDCR[FMC_Bank] = 0x000002D0;
.\driver\lib\src\stm32f4xx_fmc.c(868): error: #20: identifier "FMC_Bank5_6" is undefined
FMC_Bank5_6->SDCR[FMC_SDRAMInitStruct->FMC_Bank] = tmp1;
.\driver\lib\src\stm32f4xx_fmc.c(876): error: #20: identifier "FMC_Bank5_6" is undefined
FMC_Bank5_6->SDCR[FMC_Bank1_SDRAM] = tmp3;
.\driver\lib\src\stm32f4xx_fmc.c: 0 warnings, 30 errors
compiling stm32f4xx_dma.c...
compiling stm32f4xx_dma2d.c...
compiling stm32f4xx_gpio.c...
```

12. 这时候编译还报错,说找不到main.h, 需要删除这句话



```
26 *
27 *****
28 */
29
30 /* Includes -----*/
31 #include "stm32f4xx_it.h"
32 #include "main.h"
33
34 /** @addtogroup Template_Project
35 * @{
36 */
37
38 /* Private typedef -----*/
39 /* Private define -----*/
40 /* Private macro -----*/
```

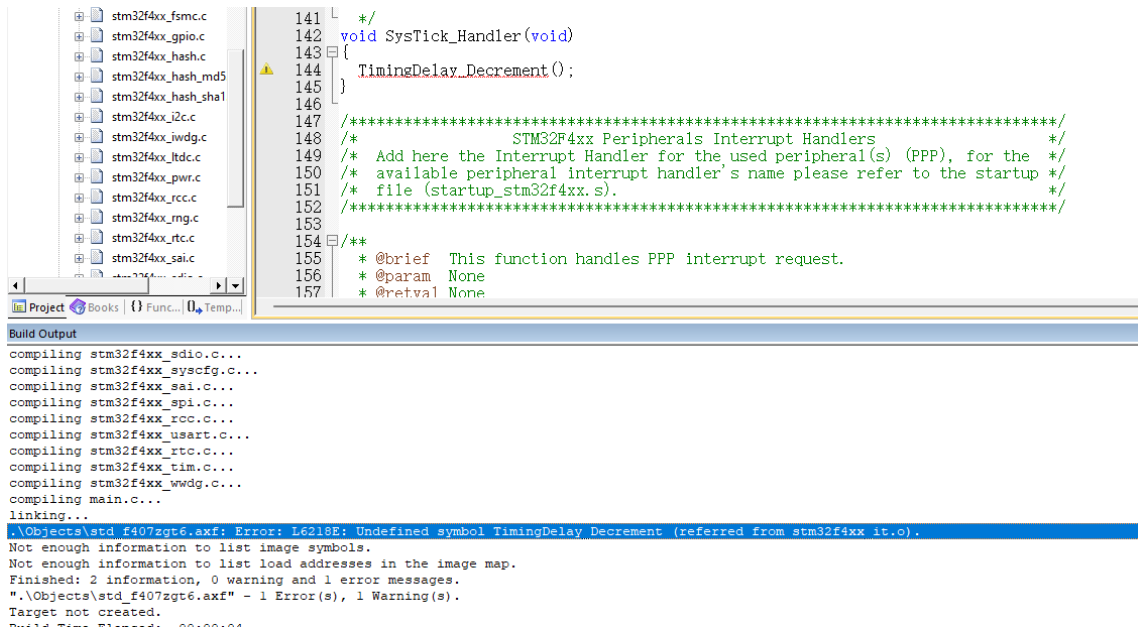
```
Build Output
Rebuild started: Project: std_f407zgt6
*** Using Compiler 'V5.06 update 7 (build 960)', folder: 'D:\develop\keil\ARM\ARMCC\Bin'
Rebuild target 'Target 1'
assembling startup_stm32f40_4lxxx.s...
compiling stm32f4xx_it.c...
.\driver\lib\src\stm32f4xx_it.c(32): error: #5: cannot open source input file "main.h": No such file or directory
#include "main.h"
.\driver\lib\src\stm32f4xx_it.c: 0 warnings, 1 error
compiling system_stm32f4xx.c...
compiling stm32f4xx_crc.c...
compiling stm32f4xx_dac.c...
compiling stm32f4xx_cryp_des.c...
compiling stm32f4xx_dbgmcu.c...
compiling stm32f4xx_cryp_tdes.c...
compiling stm32f4xx_can.c...
```



```
24 * See the License for the specific language governing p
25 * limitations under the License.
26 *
27 *****
28 */
29
30 /* Includes -----*/
31 #include "stm32f4xx_it.h"
32
33
34 /** @addtogroup Template_Project
35 * @{
36 */
37
38 /* Private typedef -----*/
39 /* Private define -----*/
40 /* Private macro -----*/
```

Figure 2

13. 再编译还报错,需要删除中断里的这个函数,因为这是ST官方例程里的,我们不需要



```
141  /*
142  void SysTick_Handler(void)
143  {
144      TimingDelay_Decrement();
145  }
146
147  /*****
148  /*      STM32F4xx Peripherals Interrupt Handlers
149  /*  Add here the Interrupt Handler for the used peripheral(s) (PPP), for the
150  /*  available peripheral interrupt handler's name please refer to the startup
151  /*  file (startup_stm32f4xx.s).
152  /*
153  /****
154  /**
155  * @brief This function handles PPP interrupt request.
156  * @param None
157  * @retval None
```

Build Output

```
compiling stm32f4xx_sdio.c...
compiling stm32f4xx_syscfg.c...
compiling stm32f4xx_sai.c...
compiling stm32f4xx_spi.c...
compiling stm32f4xx_rcc.c...
compiling stm32f4xx_uart.c...
compiling stm32f4xx_rtc.c...
compiling stm32f4xx_tim.c...
compiling stm32f4xx_wwdg.c...
compiling main.c...
linking...
.\Objects\std_f407zgt6.axf: Error: L6218E: Undefined symbol TimingDelay_Decrement (referred from stm32f4xx_it.o).
Not enough information to list image symbols.
Not enough information to list load addresses in the image map.
Finished: 2 information, 0 warning and 1 error messages.
".\Objects\std_f407zgt6.axf" - 1 Error(s), 1 Warning(s).
Target not created.
```

14. 这时候编译,在只有main.c文件的情况下,应该是0 warning,0 error